

Opposing CoC moment during contact

Opposing moment

Opposing shift → hindered alignment

Comparison with the supporting CoC case



Null-space torques and projector

Null-space projector:

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Secondary joint torques $\rightarrow$ kinematic null space

J : joint $\rightarrow$ (EE) velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

Damping torque:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity

Zero at zero joint velocity

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

Conditioning torque:

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning

Can act at zero joint velocity

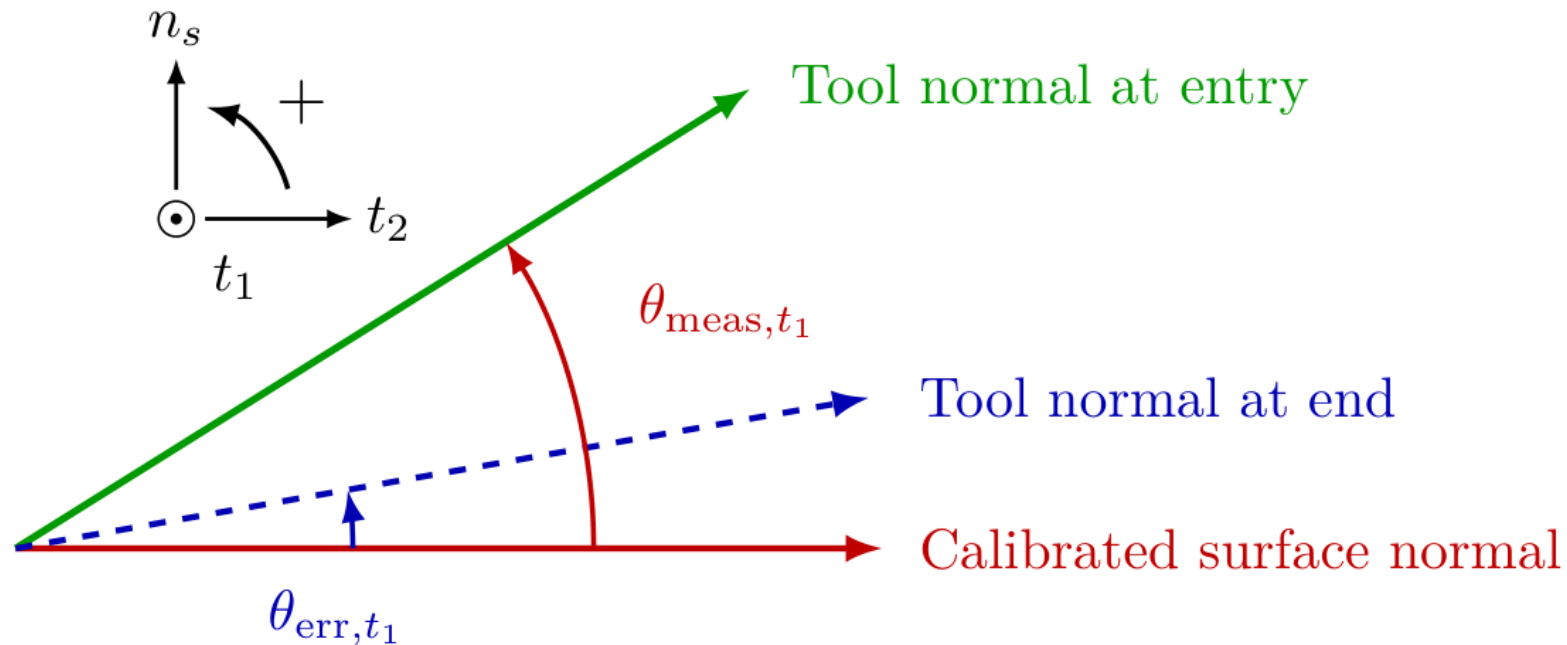
k_σ : torque magnitude [N m]

v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

Entry offset and remaining angular error

Entry offset and remaining error: relative to the calibrated surface



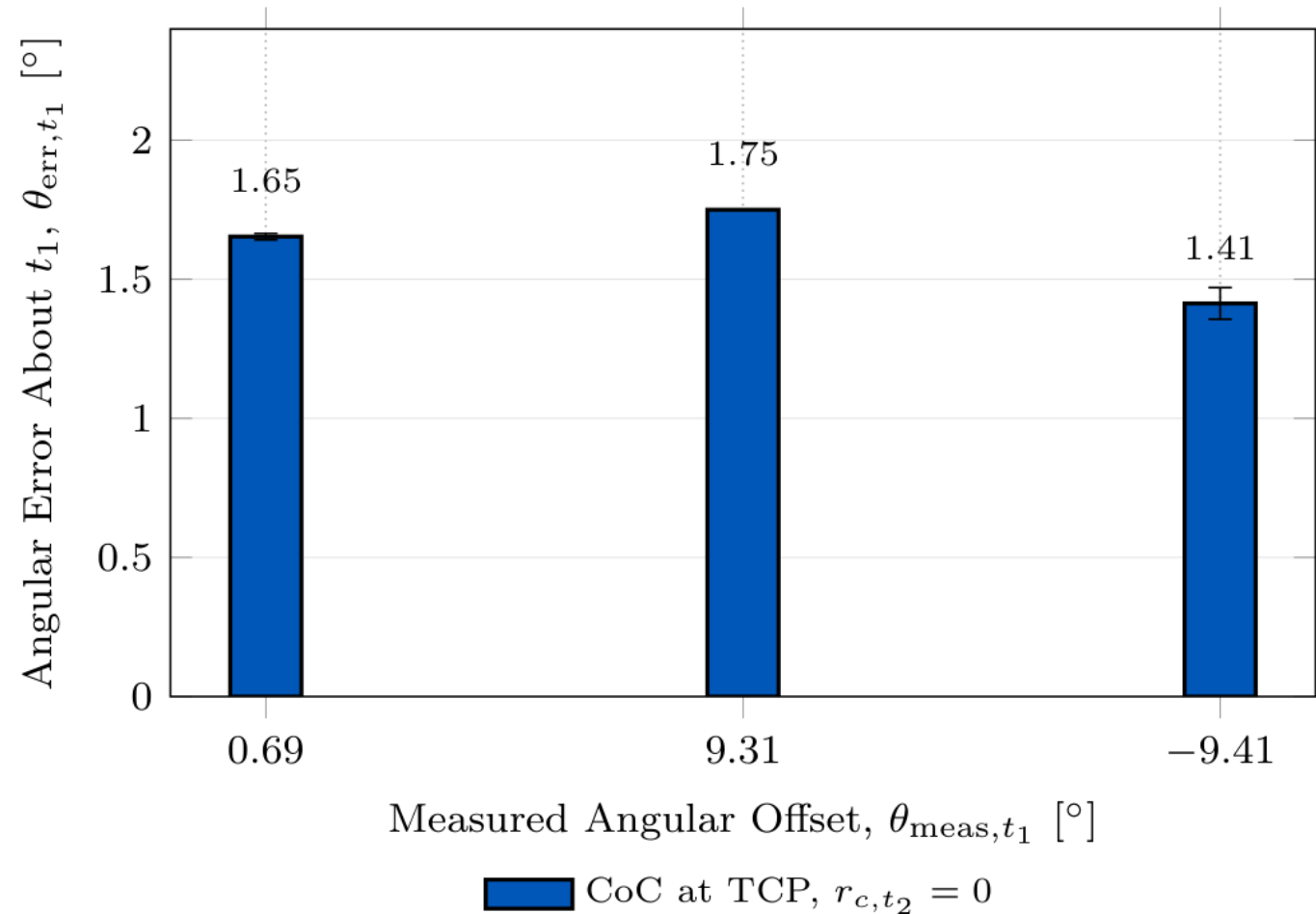
Plotted error: component about the first surface tangent

Angular error with CoC at the TCP

CoC at the TCP

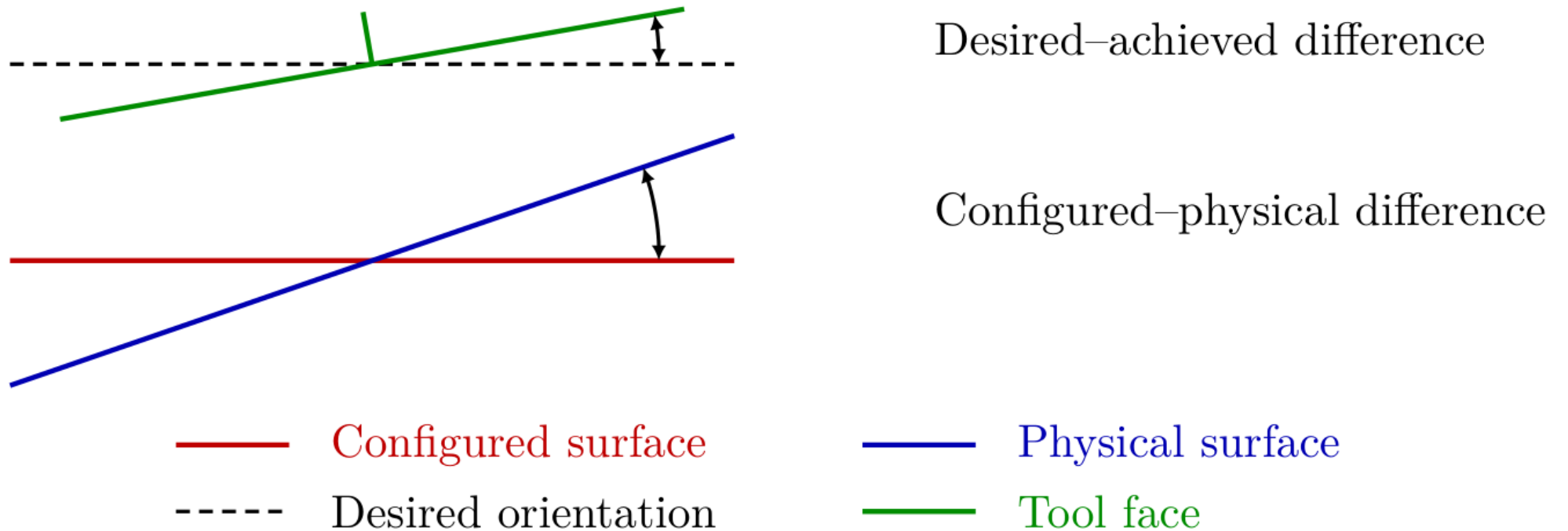
Small angular errors remain after contact

Reduced starting tilt
Residual angular error



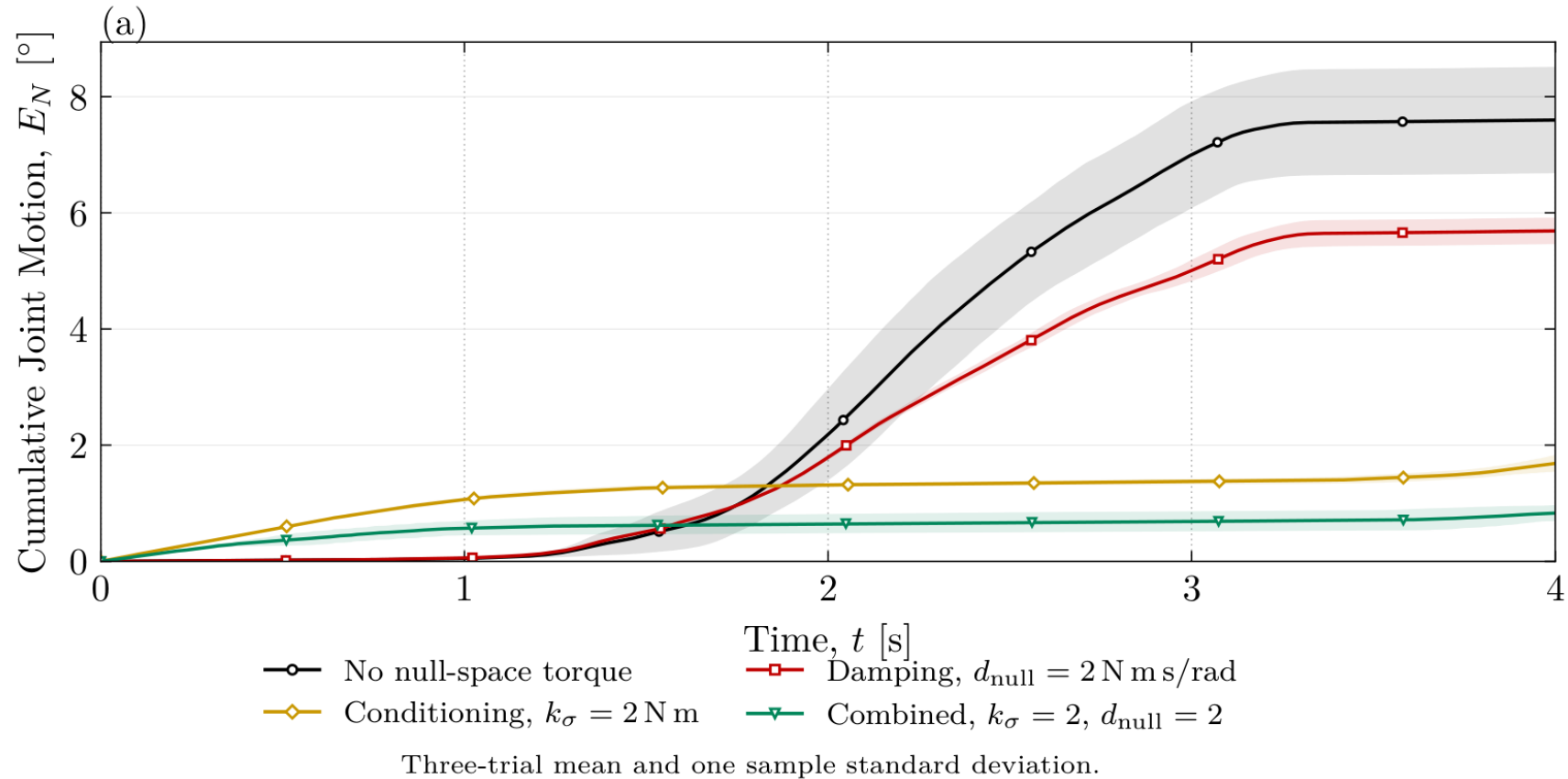
Sources of angular offset

Starting tilt: surface-calibration + tool-orientation differences



Both differences contribute to the entry angle

Damping reduces cumulative joint motion



≈25% less cumulative motion with damping alone

≈50% less cumulative motion with damping added to conditioning